

Design for Maintainability

Case Study with Aviation Accident ENG12IA010

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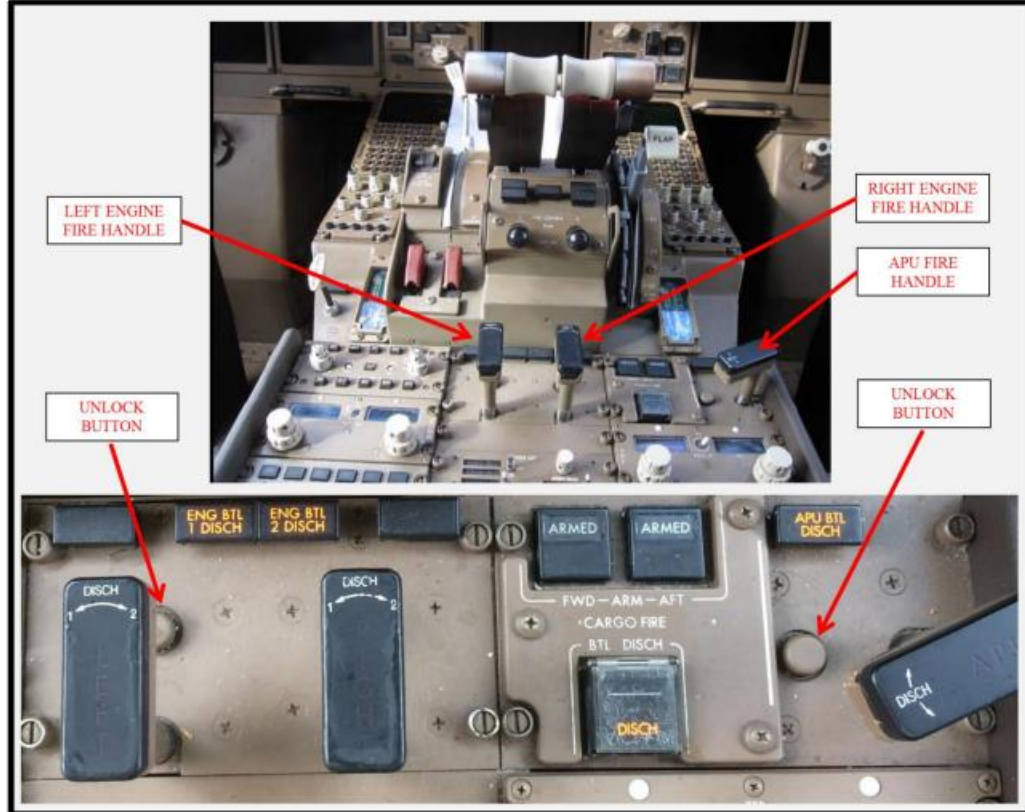
Aviation Accident Summary

American Airlines Boeing 767 Engine Fire – February 8, 2012



- An undercowl engine fire occurred in the right engine shortly after takeoff from John F. Kennedy International Airport (JFK) in Jamaica, New York.
- As the airplane was climbing through 9,000 feet, the pilots heard a bang that was immediately followed by a right engine fire warning in the cockpit.

Figure 11 Photograph showing the engine fire control panel.



The pilots shutdown the right engine and discharged both fire bottles into the right engine's nacelle before the fire warning was extinguished...



...and declared an emergency and returned to JFK for a single-engine, overweight landing without further incident.

NSTB Probable Cause & Findings

Maintainer Error.

The **failure of the American Airlines aviation maintenance technicians** to properly reassemble the two-piece bracket and spray shield... that caused a fuel leak that sprayed out and ignited on hot engine cases.

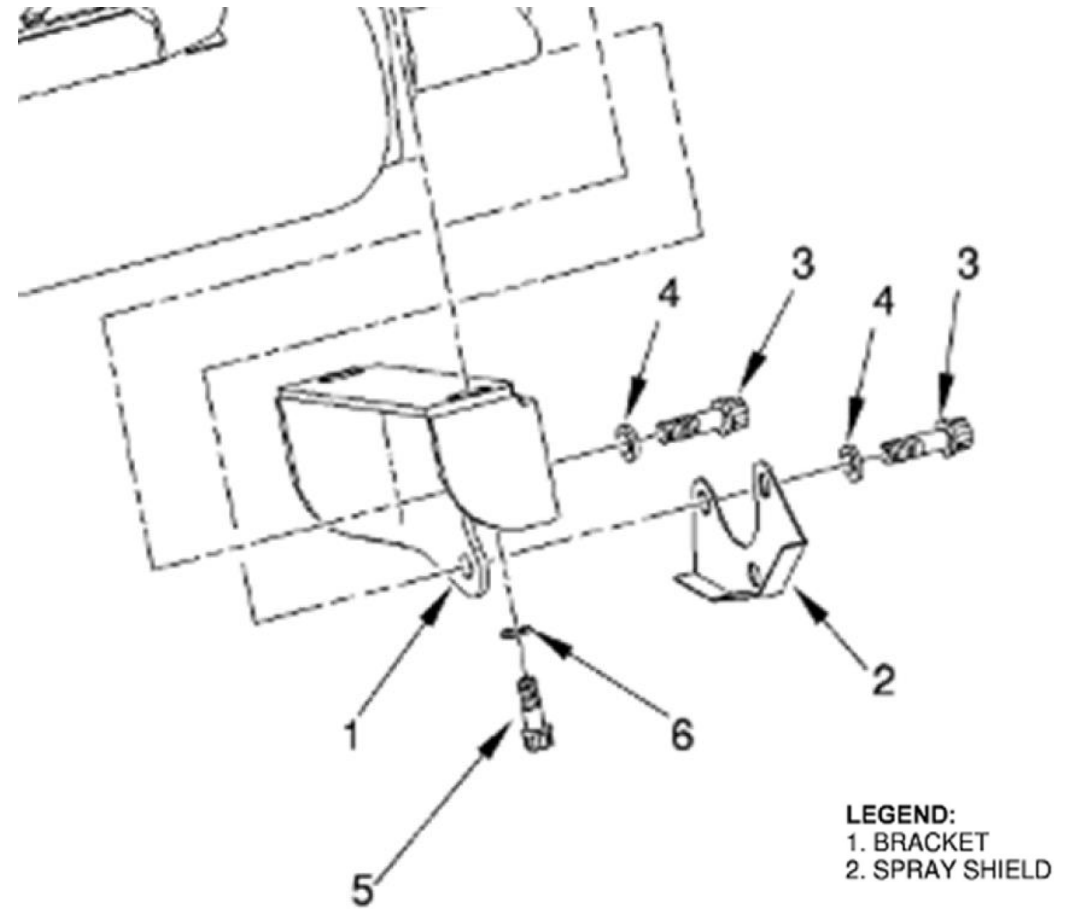
Workers don't cause failures.

Worker's “trigger” latent conditions that lie dormant in our organizations.

Errors are consequences not causes.

Could the correct configuration be more intuitive and/or better communicated?

The NTSB also noted that the B767 Aircraft Maintenance Manual's lack of pictorial displays of the correct assembly of the two-piece bracket and spray shield contributed to the cause of the fire.



NSTB Probable Cause & Findings

Maintainer Error.

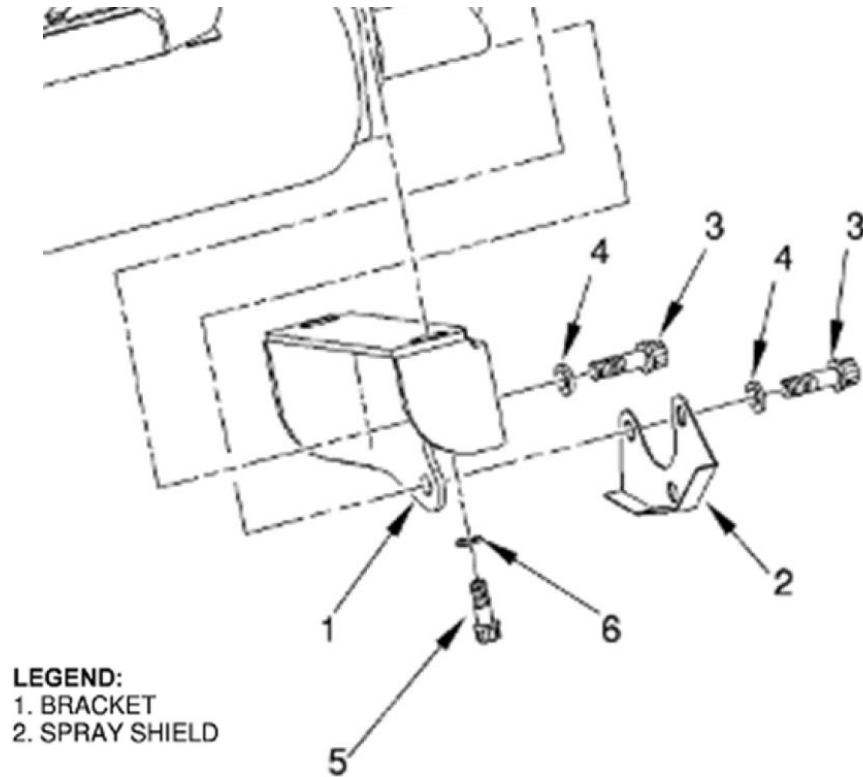
The **failure of the American Airlines aviation maintenance technicians** to properly reassemble the two-piece bracket and spray shield...exchanger that caused a fuel leak that sprayed out and ignited on hot engine cases.

Contributing to the cause of the fire was the 767 Aircraft Maintenance Manual's lack of any graphical or pictorial displays of the correct assembly of the two-piece bracket and spray shield.

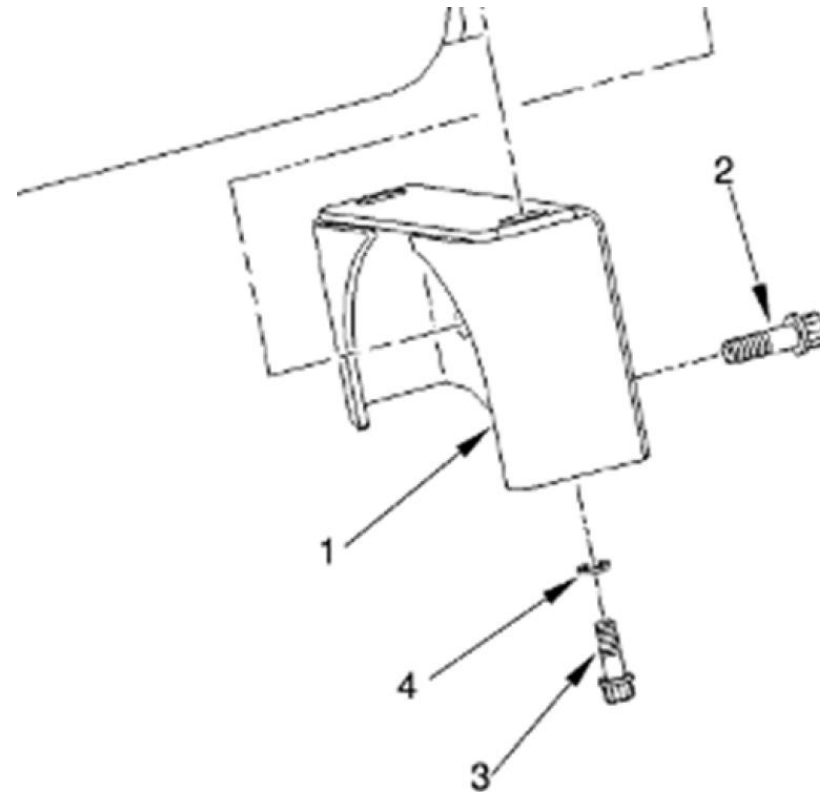
Could this be designed to be impossible?

A [service bulletin](#) was later published calling for utilizing a new simplified one-piece supporting bracket.

Old Configuration: Prone to Misassembly



New Configuration: Simplified



SB 73-A0401 R00 ENGINE FUEL AND CONTROL – GENERAL (73-00-00) - IDG FUEL-OIL COOLER FORWARD FLANGE BRACKET AND SPRAY SHIELD REPLACEMENT



*“If you define the problem
correctly, you almost have
the solution.”*

(Steve Jobs)

NSTB Probable Cause & Findings

Maintainer Error.

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MIL-STD-40051-1D

4.9.4.1 Warning <warning>.

A warning shall be used to identify a clear danger for injury or death to the person doing that procedure. A warning shall also be used when there is both danger of injury or death to personnel and danger of damage to the equipment during that procedure.

4.9.4.2 Caution <caution>.

A caution shall be used to identify a clear risk of damage to the equipment if the procedure is not followed correctly.

CAUTION: Misintallation of bracket and spray shield orientation can cause fuel leaks and engine fires.

NSTB Probable Cause & Findings

Maintainer Error.

The **failure of the American Airlines aviation maintenance technicians** to properly reassemble the two-piece bracket and spray shield...exchanger that caused a fuel leak that sprayed out and ignited on hot engine cases.

Contributing to the cause of the fire was the 767 Aircraft Maintenance Manual's lack of any graphical or pictorial displays of the correct assembly of the two-piece bracket and spray shield.

(1) First task step description

(2) Second task step description

CAUTION: Misintallation of bracket and spray shield orientation can cause fuel leaks and engine fires. Figure # illustrates proper assembly.

(3) Install bracket and spray shield (Figure #) as follows:

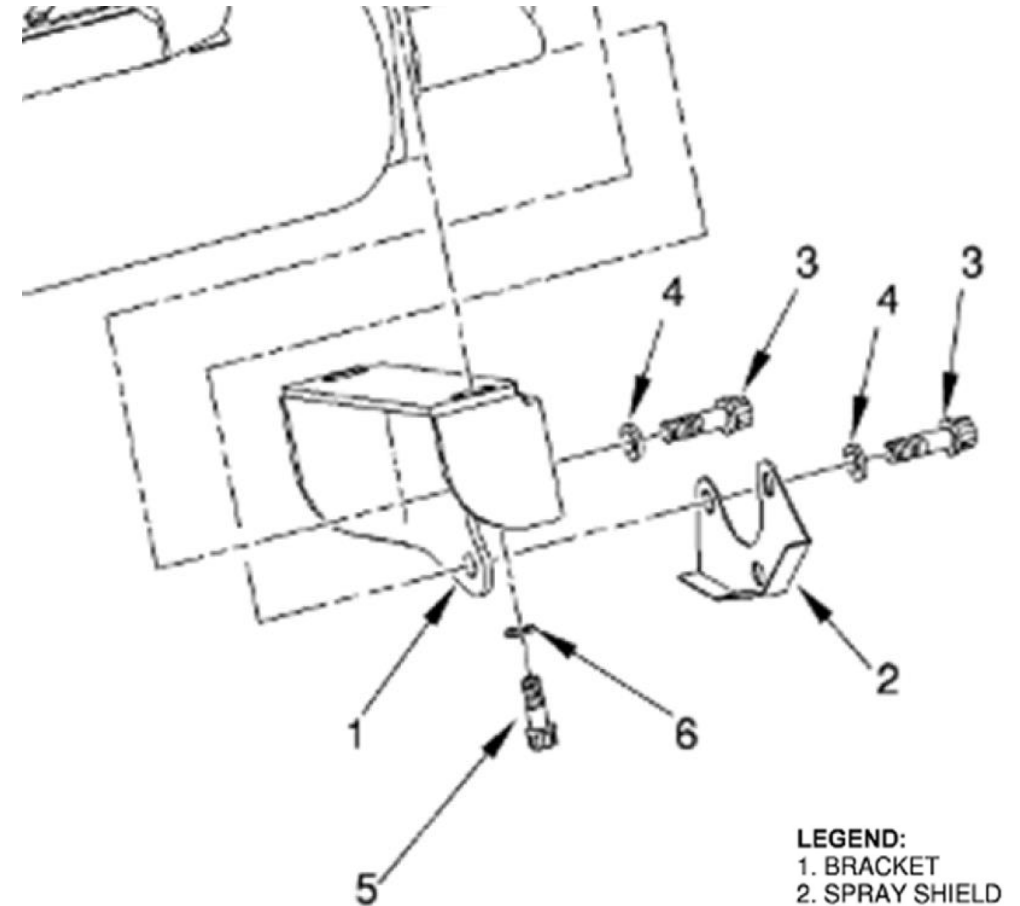
(a) substep 1, such as lubrication

(b) substep 2, etc.

(install washers, torque bolts, etc)

(4) Subsequent task description

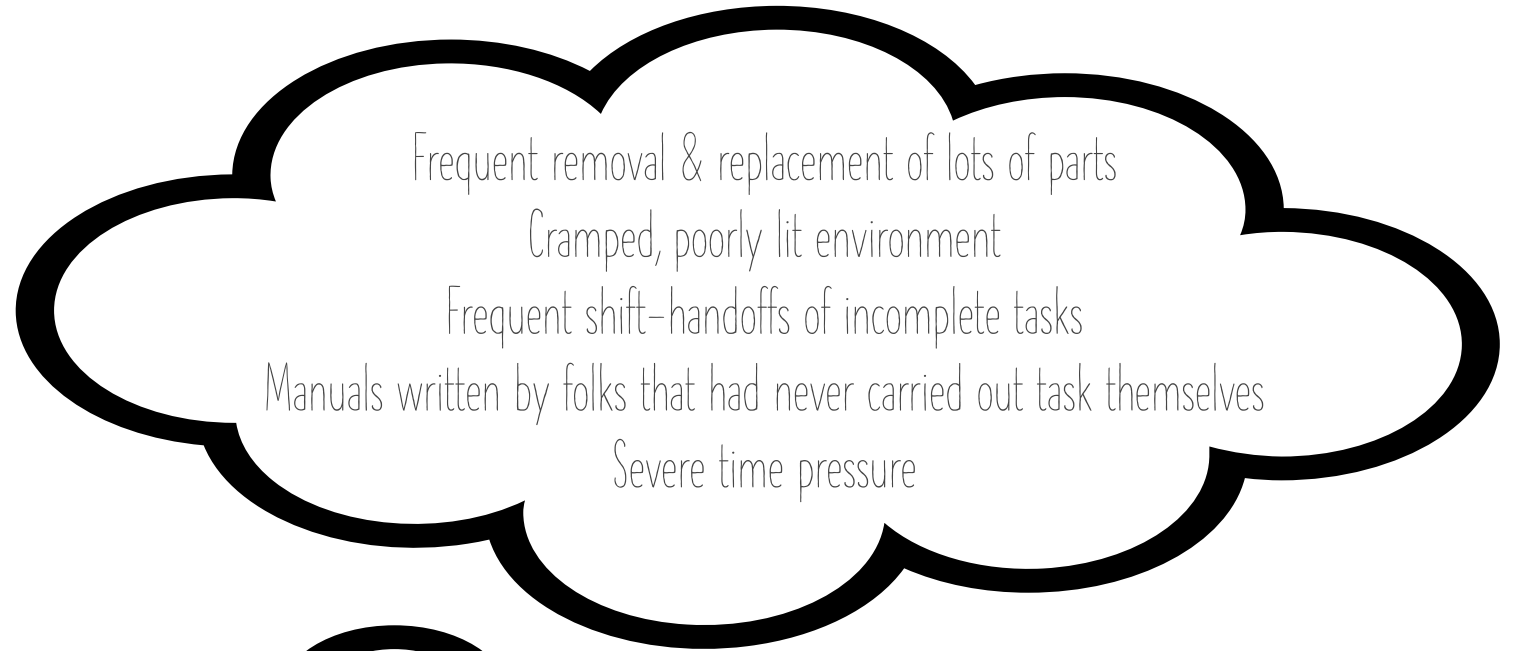
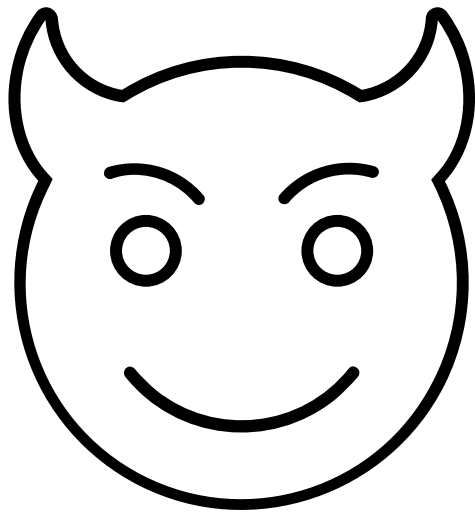
Figure # - likely shown a couple pages later following the text...



Another thing to note....

- Bracket/Spray shield did not need any maintenance work. They had to be removed in order to access another component.
- Consider designs that have frequently removed and replaced components as close to exterior as possible.
- Removal and (more so!) *Replacement* are very error prone tasks!

How could an evil genius create an activity destined to produce an abundance of human errors?



Frequent removal & replacement of lots of parts

Cramped, poorly lit environment

Frequent shift-handoffs of incomplete tasks

Manuals written by folks that had never carried out task themselves

Severe time pressure

Maintenance tends to be a highly error-productive activity

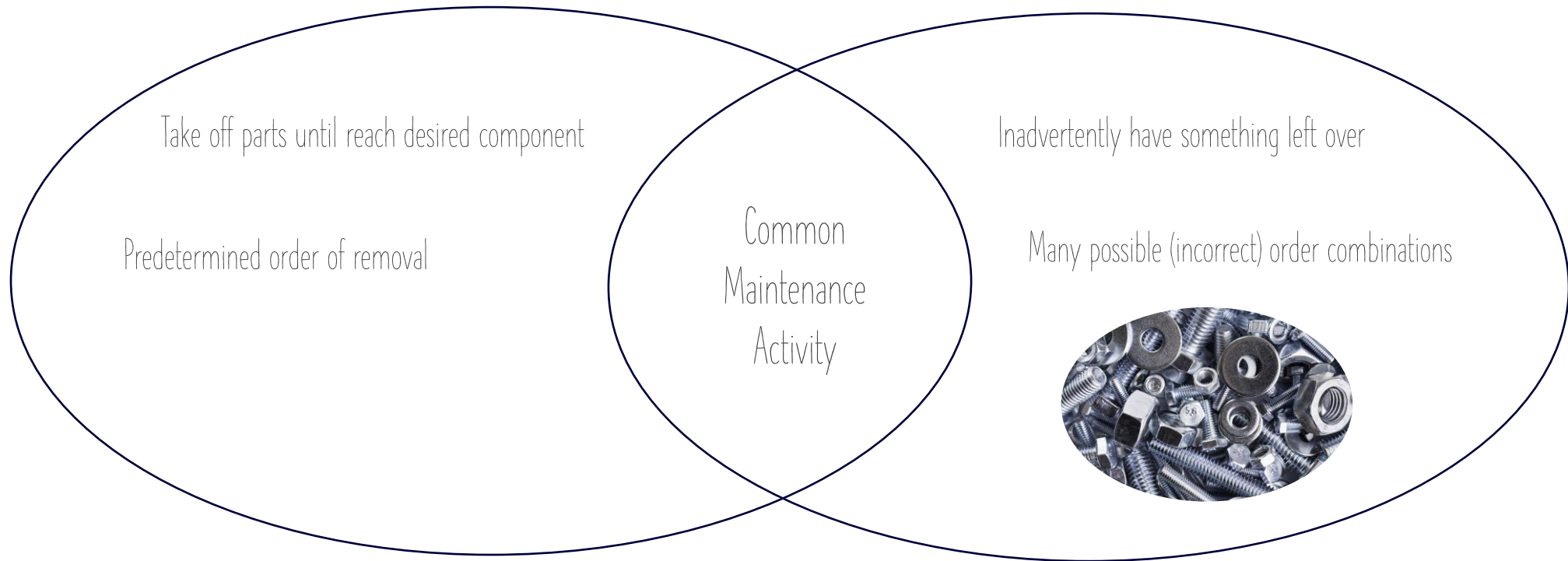
Maintenance errors are not random.

They fall into **systemic patterns** relating to both the nature of the activity and types of errors involved.

(Reason/Hobbs)

Removal

Replacement / Installation



Reassembly & Installation associated with large majority of errors made

Error of Commission

doing something, but
the wrong thing

Incorrect part used
Assembly order incorrect
Incorrect torque applied

Error of Omission

not doing something,
skipping the right thing

Fastenings left incomplete/undone
Items left locked, loose, or disconnected
Items missing
Tools/spare fastenings not removed
Lack of lubrication
Panels left off

Omissions = largest single category of maintenance errors

Management of Omissions

Step 1: Identify Omission-Prone Steps

Step 2: Provide Suitable Reminders

Omission-Provoking Features:



Memory Loading

- More likely to omit if a lot of info needs to be remembered at a time



Conspicuity

- Concealed steps at risk to be skipped



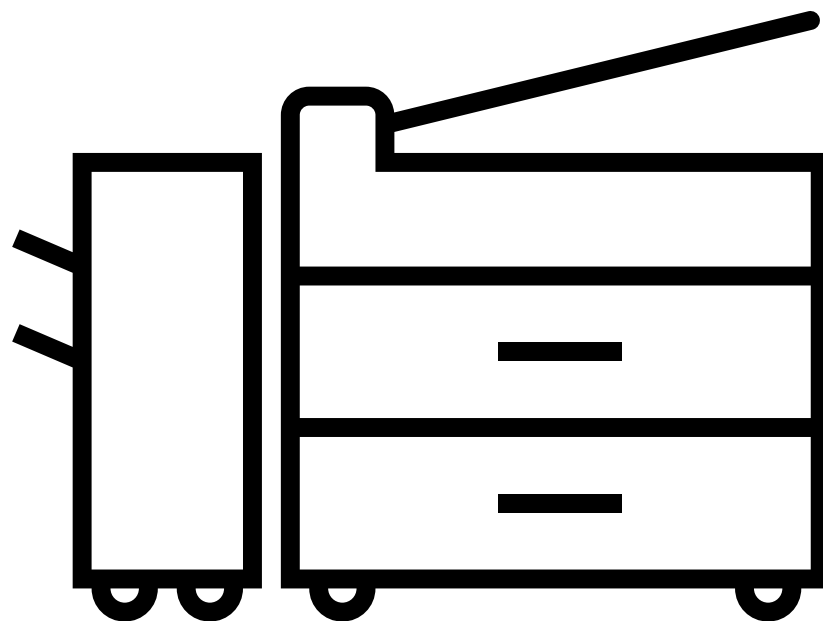
Position

- Steps toward end of task subject to premature exit risk



Local Cueing

- Functionally isolated steps more likely to be missed



Management of Omissions

Step 1: Identify Omission-Prone Steps

Step 2: Provide Suitable Reminders

Characteristics of a Good Reminder:



Conspicuous

- Grab attention at the critical time



Close

- As close as possible to relevant step



Context

- When and Where to execute the step



Content

- Clearly notes *what* has to be done



Count

- Ability to count off number of discrete actions

Managing Maintenance Error

A Practical Guide



An Ashgate Book

James Reason and Alan Hobbs

Situations and systems are easier to change than the human condition - particularly when people are well-trained and well-motivated, as they usually are in maintenance organisations.

This is a down-to-earth practitioner's guide to managing maintenance error, written in Dr. Reason's highly readable style. It deals with human risks generally and the special human performance problems arising in maintenance, as well as providing an engineer's guide for their understanding and the solution.

After reviewing the types of error and violation and the conditions that provoke them, the author sets out the broader picture, illustrated by examples of three system failures. Central to the book is a comprehensive review of error management, followed by chapters on:- managing person, the task and the team; - the workplace and the organization; - creating a safe culture; It is then rounded off and brought together, in such a way as to be readily applicable for those who can make it work, to achieve a greater and more consistent level of safety in maintenance activities.

The readership will include maintenance engineering staff and safety officers and all those in responsible roles in critical and systems-reliant environments, including transportation, nuclear and conventional power, extractive and other chemical processing and manufacturing industries and medicine.